

Wavelet and Adaptive Coordinate Attention Guided Fine-Grained Residual Network for Image Denoising

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Abstract—Convolutional neural networks (CNN) have achieved remarkable performance in image denoising. However, most existing CNNs cannot accurately capture and remove tiny noises during the denoising process and lose edge detail information easily. In this paper, we propose a fine-grained residual network guided by wavelet and adaptive coordinate attention (WACAFRN) for image denoising. Firstly, we propose an adaptive coordinate attention mechanism and combine it with cascaded Res2Net residual blocks to form an encoder network for more accurate noise removal. Secondly, we propose a wavelet attention mechanism that combines global and local residual blocks to form a decoder network, aiming to address the problem of edge detail information loss. At last, we complement the noise information through a noise estimation block to further enhance the model's ability to adapt to noise. Extensive experiment results demonstrate that our proposed method outperforms existing denoising methods in both qualitative and quantitative aspects. Notably, our method significantly improves real-world noise removal tasks on the CC dataset, with an average increase of 2.08 dB in PSNR and 0.0264 in SSIM over the state-of-the-art methods. Additionally, WACAFRN exhibits faster inference speeds, underscoring its efficiency in real-world applications.

Index Terms—Image denoising, wavelet attention, adaptive coordinate attention, noise estimation, residual connection.

I. INTRODUCTION

IMAGE denoising is one of the key tasks in the field of computer vision and is of vital significance [1], [2], [3].

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Images contain a wealth of information that can help people understand and comprehend the world around them. However, external factors such as harsh environmental conditions, technical defects in the image capture process, and human factors can lead to image quality degradation and destruction, which in turn affects people's acquisition and use of the information. Hence, removing noise to obtain high-quality images is a significant and ongoing challenge [4].

In the past few decades, significant advancements have been made in the field of image denoising research [5], [6]. Traditional image denoising methods mainly include spatial domain filtering methods [7], frequency domain filtering methods [8], methods based on Partial Differential Equations (PDE) [9], and non-local self-similar methods [10]. These methods are effective in noise removal and image restoration, but the computational complexity is high, and the process of adjusting parameters is difficult.

Some specific algorithms, such as Non-Local Means (NLM) [11], Weighted Nuclear Norm Minimization (WNNM) [12], and Block-Matching and 3D Filtering (BM3D) [13], all use similar strategies to find and utilize similar structures in images. These methods can better preserve the structure and detailed information of the image, but the computational complexity is high, which may pose challenges to application scenarios with high real-time requirements.

The powerful learning capabilities of deep learning models present a particularly promising research area for image denoising. Jian et al. [14] demonstrate the effectiveness of gradient-based convolutional networks in restoring noisy and degraded images. The Multi-layer Perceptron (MLP) proposed by Burger et al. [15] can achieve a good noise reduction effect but the receptive field is limited. Schmidt et al. [16] proposed the Shrink Field Cascade (CSF) model to improve computational efficiency and image restoration quality. Chen et al. [17] proposed the TNRD model, which can be adapted to various restoration tasks. Daniel et al. [18] used the Gaussian mixture model to analyze image blocks and proposed the EPLL denoising method. Zhang et al. [19] proposed DnCNN, the first deep CNN-based Gaussian denoising method, which remains an important benchmark for the design of image denoising methods due to its excellent design and the concept of residual learning. Building on this work, Zhang et al. [20] proposed FFDNet, which utilizes noise level maps and noisy images to improve adaptability to noise. Subsequently, Zhang et al. [21]

also developed IRCNN, a set of CNN-denoising networks for image denoising.

RIDNet [22] is a denoising method that combines feature attention and residual connection, which has great advantages in real image denoising tasks but has the disadvantage of over-smoothing. Tian et al. [23] proposed ADNet, which uses a dilated convolution-based sparse mechanism and an attention mechanism. Later, they proposed a dual CNN [24] for image denoising. Ma et al. [25] introduced the deep boosting strategy to the field of image denoising, which achieved advanced results. Jiang et al. [26] proposed an adaptive prior image denoising method, which improved the image denoising effect by introducing two regularization priors and achieved excellent performance. Zamir et al. [27] proposed an efficient transformer model for high-resolution image restoration, achieving outstanding results on multiple restoration tasks. Tian et al. [28] proposed a multi-stage image denoising method combined with wavelet transform to speed up training and improve denoising performance.

Although researchers have achieved superior image denoising effects using deep convolutional neural networks, existing methods still struggle with accurately locating and learning noise during the denoising process. This limitation often leads to the loss of edge information in the denoising results. To overcome the above problems, in this paper, we propose a fine-grained residual denoising network guided by wavelet and adaptive coordinate attention. Specifically, we propose an enhancement to deep denoising networks, integrating an adaptive coordinate attention mechanism with stacked fine-grained residual blocks in the encoder for better noise learning, and a wavelet attention mechanism in the decoder layer for improved image edge restoration. Additionally, we introduce a noise estimation sub-network for prior knowledge guidance and a feature fusion block for integrating the estimated noise with the channel noise map for refined noise mapping. Our experimental results demonstrate that our method, named WACAFRN, effectively enhances network denoising abilities and the quality of restored images. To sum up, this paper mainly has the following contributions:

- To more accurately capture and learn fine noise points, we propose an adaptive coordinate attention mechanism (ACA), which is combined with stacked fine-grained residual blocks to form an encoder layer. ACA guidance can adaptively learn attention intensities to accurately extract noise information at different encoder layers.
- To preserve image edge information more effectively and improve the quality of image restoration, we propose a wavelet attention mechanism (WA), which is combined with global and local residual connections to form a decoder layer. WA guidance helps preserve detailed edge information in images.
- Based on introducing ACA at the encoder layer and WA at the decoder layer respectively, we design a noise estimation block (NEB) to complement the noise prior information, and by combining the encoder-decoder backbone network and the noise estimation block, we propose

the wavelet and adaptive coordinate attention guided fine-grained residual denoising network.

- Extensive experiments demonstrate that our method significantly improves the denoising ability of the network and the quality of restored images.

The remainder of this paper is organized as follows. Section II presents some work related to our method. We detail our proposed WACAFRN in Section III. In Section IV, we organize extensive experiments to demonstrate the effectiveness of our proposed method and compare it with the existing state-of-the-art methods. Finally, in Section V, we provide a summary and outlook of our work.

II. RELATED WORK

A. Coordinate Attention

Coordinate Attention [29] is an attention mechanism for computer vision tasks that introduces positional spatial information on feature maps such that attention weights depend not only on feature similarity within a channel but also on feature differences between different positions. Coordinate Attention has achieved excellent performance in many computer vision tasks, especially in object detection, image segmentation, and human pose estimation. These facts clearly show that the power and efficiency of Coordinate Attention in image tasks are very competitive. Based on this reality, we propose an adaptive coordinate attention mechanism for image denoising to capture and remove fine noise more accurately.

B. Wavelet Transform

In image denoising, the wavelet transform [30] is used to remove noise from an image by thresholding the wavelet coefficients in high-frequency subbands. In feature extraction, wavelet transform is used to capture the multi-scale and multi-directional features of images, which can be used for further image analysis. Liu et al. [31] proposed a multilevel wavelet-CNN (MWCNN) architecture for image restoration, which combines the invertibility, frequency, and location properties of the wavelet transform to help image restoration. Tian et al. [28] proposed a multi-stage image denoising algorithm combined with wavelet transform to enhance the feature information extracted from the original CNN block and use the wavelet transform to improve the denoising effect. Huang et al. [32] proposed a wavelet-inspired invertible network (WINNet) aimed at combining the advantages of wavelet transformation and learning-based methods, while simultaneously enhancing interpretability. Inspired by the above, we introduced wavelet transformation in the decoding stage. Unlike the previous approach, we devised a more effective wavelet attention mechanism.

C. Res2Net

Res2Net is a convolutional neural network architecture for image recognition [33]. Different from traditional convolutional neural networks, each ResBlock in Res2Net contains multiple branches with different feature map sizes but shares the same number of feature map channels. This design enables

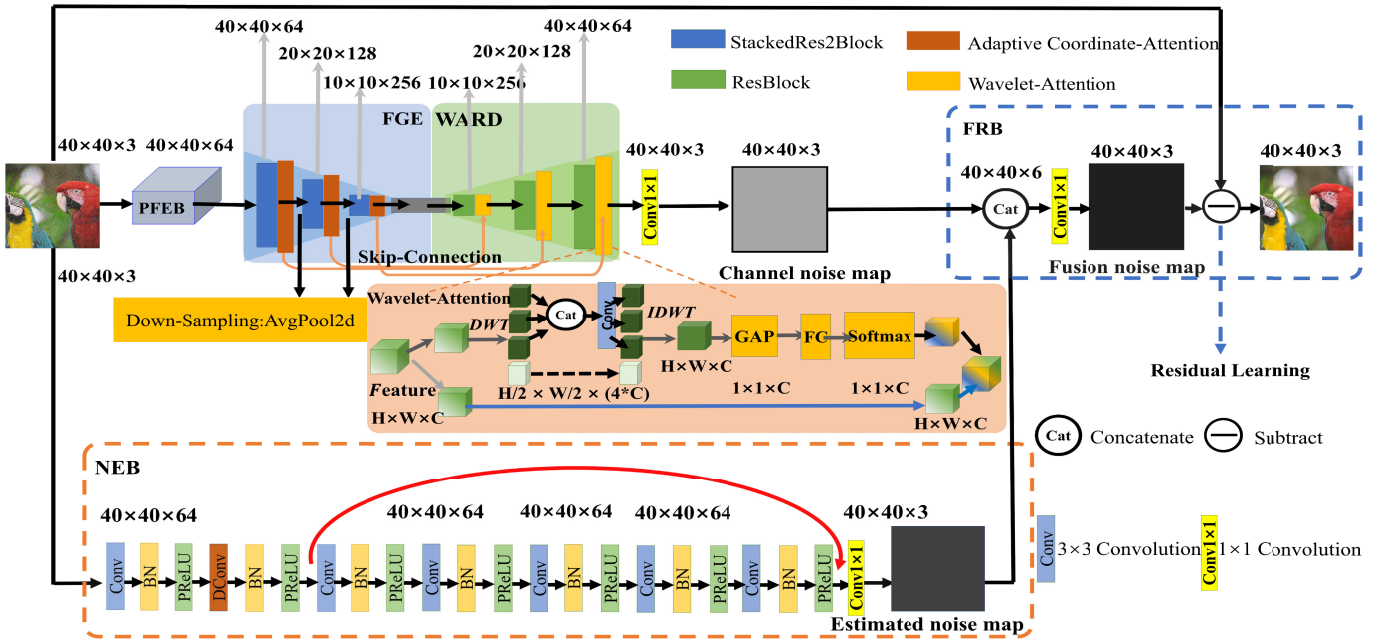


Fig. 1. The architecture of WACAFRN proposed for image denoising. WACAFRN consists of five components: PFEB, FGE, WARD, NEB, and FRB. Noisy images are initially parallelly fed into the PFEB and NEB blocks. After processing through the PFEB block, basic features are obtained, and further channel feature maps are extracted through the FGE and reconstructed using WARD. The branch processed by the NEB block corresponds to the generation of estimated noise maps. Finally, the FRB fuses the features from both branches and achieves denoising through residual learning.

Res2Net to make full use of feature information of different scales and levels. Encouraged by these facts, in this study, we propose a fine-grained feature extraction network by stacking Res2Net blocks and combining them with an adaptive coordinate attention mechanism for image denoising tasks.

D. Noise Estimation

Noise estimation is a very important technique in deep learning-based image denoising [34]. The advantage of using noise estimation is that the parameters of the denoising model can be adaptively adjusted according to the different noise levels of the image. The method adopted by FFDNet [20] is to combine the global noise map and the noisy subgraph as input, which is a way of combination in the early stage. The noise estimation subnetwork and denoising network in CBDNet [35] are jointly trained. It also combines the noise map and the noisy input as the input of the denoising network. The recently proposed denoising diffusion models, DDPM [36] and Repaint [37], have the capability to achieve advanced noise estimation results. However, they require a significant number of computational resources and have large model sizes. Inspired by these works, we propose a noise estimation block to estimate noise information, which is then fused with the output of the backbone network to improve denoising performance. It's important to note that, considering factors such as computational cost and model size, we have not employed denoising diffusion models for noise estimation. Instead, noise estimation serves as a supplementary role, aiming for lightweight and easy-to-train solutions.

III. THE PROPOSED METHOD

In this part, we describe our proposed method in detail. WACAFRN mainly consists of five parts, which we explain

one by one in each subsection. First, in Section A, we introduce the overall framework of our model. Then, we describe the proposed primary feature extraction block (PFEB) in detail in Section B. Section C introduces the fine-grained encoder (FGE). A detailed introduction to the wavelet attention residual decoder (WARD) is in Section D. Section E introduces the NEB. Finally, Section F introduces the fusion reconstruction block (FRB).

A. Network Architecture

The WACAFRN proposed in this study consists of five parts: PFEB, FGE, WARD, NEB, and FRB as shown in Fig. 1. The PFEB block aims to increase the receptive field and feature extraction range of the network and provide sufficient shallow feature information for subsequent modules. It should be noted that WACAFRN is a two-branch convolutional neural network. In addition to being input to PFEB, the noisy image will also be passed to NEB in parallel to learn and extract global noise estimation information.

The backbone of the model is an asymmetric encoder-decoder structure designed to extract and learn the main channel noise feature maps. The encoding part is a fine-grained encoder, which achieves fine feature extraction by stacking Res2Net blocks and combining adaptive coordinate attention mechanism, which can capture and learn fine noise information that is difficult to obtain by traditional convolutional neural networks. During the encoding process, as we incorporate down-sampling, the image's resolution undergoes a gradual reduction, resulting in a more concentrated and crucial extraction of feature information. The decoding part is a wavelet attention residual decoder, which combines the global local residual connection and the wavelet attention mechanism and weights the channels through the features of the frequency

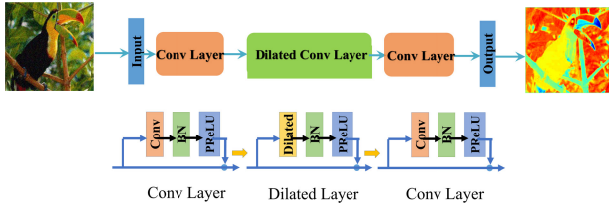


Fig. 2. Detailed structural illustration of PFEB. PFEB consists of three convolution/dilated convolution layers designed for extracting initial features.

domain, which can reduce the information loss while retaining the edge details of the image. At the same time, the quality of image restoration can be improved. NEB can obtain a global noise map by modeling and learning noisy images, which will provide more information about noise distribution and noise categories for the final noise map. FRB uses a residual learning strategy to fuse channel noise maps and noise estimate maps and combine them to get the final merged noise map. Finally, a clean image is obtained by subtracting the combined noise map from the model's input (the noisy image).

B. Primary Feature Extraction Block

The first part of WACAFRN is the primary feature extraction block, used to provide initial shallow-level feature information. The primary feature extraction block consists of three layers: a convolutional layer, a dilated convolutional layer with a dilation factor of 2, and a convolutional layer. The detailed structural illustration of PFEB is shown in Fig. 2. In Fig. 2, the input of the PFEB block is a noisy image, and the output is the extracted primary feature map. To observe the extracted features more intuitively, the output feature map is displayed in the form of thermodynamic images. By observing Fig. 2, it can be found that the primary feature map extracted by PFEB can cover most of the noise positions and provide better shallow features for the model. The realization of these can be represented by the following formula:

$$O_{PFEB} = f_{C2}(f_{D1}(f_{C1}(N))), \quad (1)$$

where N is the input noise image, f_{C1} denotes the function of the first convolutional layer, f_{D1} is the function of the second convolutional layer and is the function of the first dilated convolutional layer, and f_{C2} is the function of the second convolutional layer.

C. Fine-Grained Encoder

To effectively extract finer-grained noise information and remove as much noise as possible, we design a fine-grained encoder, which is composed of three fine-grained encoding layers (FGEL). Each FGEL contains the following operations: changing the convolution of the feature channel, a fine-grained feature extraction block composed of stacked Res2Net blocks and an adaptive coordinate attention mechanism, and down-sampling to reduce the image resolution. The processing process of FGE can be expressed as follows:

$$O_{FGE} = f_{S3}\{f_3[f_{S2}(f_2\{f_{S1}[f_1(O_{PFEB})]\})]\}, \quad (2)$$

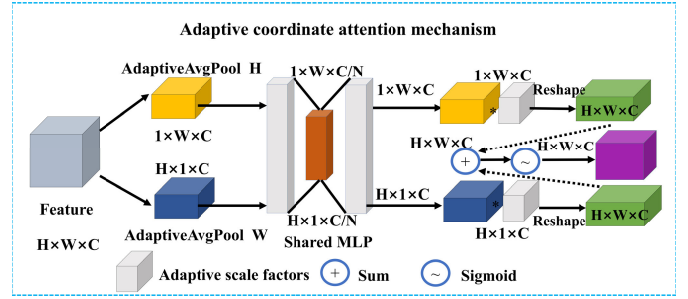


Fig. 3. Illustration of our proposed ACA module. Specifically, firstly, adaptive average pooling is applied separately to the input feature map in both horizontal and width directions. Subsequently, the results of these two adaptive average pooling operations undergo transformations such as MLP and adjustment of attention intensity factors. Finally, the results from both directions are summed, and attention weights are generated through a sigmoid function.

where O_{PFEB} is the output of PFEB, f_1 , f_2 , and f_3 are convolution functions that respectively represent the number of channels changed from 64 to 64, 64 to 128, and 128 to 256, f_{S1} , f_{S2} , and f_{S3} are fine-grained feature extraction blocks and O_{FGE} is the output of FGE. The structure of the adaptive coordinate attention mechanism is shown in Fig. 3, and the structural illustration of the fine-grained encoder is shown in Fig. 4.

To guide the fine-grained feature extraction network to better capture and extract fine noise, we propose an adaptive coordinate attention mechanism. Unlike conventional channel attention, the adaptive coordinate attention mechanism highlights important information in images by adaptively calculating attention weights. It utilizes adaptive average pooling operations to compute the mean values of the image in both the vertical and horizontal directions. These mean values are then processed through convolutional layers and activation functions. Subsequently, by adjusting adaptive scaling factors, the feature maps from the vertical and horizontal directions are weighted and combined. Finally, the attention weights are constrained between 0 and 1 using the Sigmoid function. This adaptive attention weight calculation enables the model to automatically learn important regions based on the characteristics of the image, emphasizing relevant information. Unlike the original coordinate attention mechanism, we have incorporated learnable adaptive scaling factors into the attention weight calculation, enabling the model to adapt to different types and levels of noise. By adjusting these scaling factors, the model can flexibly accommodate images with varying noise characteristics and fine-tune the degree of noise suppression as needed. The adaptive coordinate attention mechanism can be expressed as follows:

$$ACA(x) = x * \sigma(S_{FH}(M_L(\text{adaptive_avg_pool}_H(x))) + S_{FW}(M_L(\text{adaptive_avg_pool}_W(x))))), \quad (3)$$

where x represents the input feature map, $*$ represents element-wise multiplication, σ represents sigmoid activation function. M_L represents a shared MLP. S_{FH} and S_{FW} are factors for adaptively adjusting attention in the H and W directions, respectively, and can flexibly change the intensity of attention through learning.

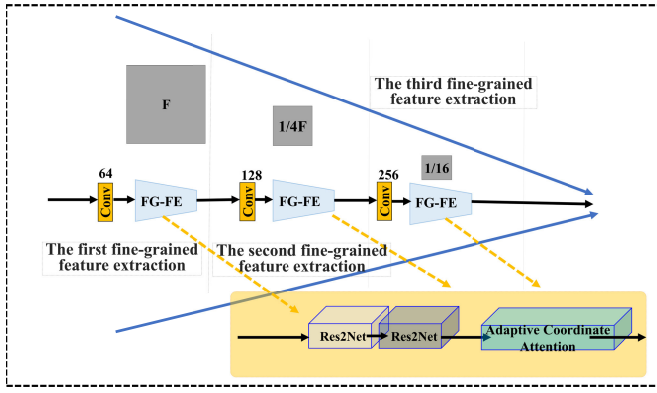


Fig. 4. Illustration of the detailed architecture of the FGE. FGE comprises three stages of fine-grained feature extraction, implemented through stacked Res2Net blocks and ACA. During this process, the resolution of the feature maps gradually decreases.

The core of the encoding layer is the fine-grained feature extraction block, which consists of stacked Res2Net blocks and an adaptive coordinate attention mechanism. Res2Net enhances feature expression ability by segmenting and reorganizing noise features of different scales, and can better capture small noise points. The stacked Res2Net block is realized by cascading two Res2Net blocks, which can further improve the representation ability of the model and extract finer-grained noise features. The combination of adaptive coordinate attention mechanism and stacked residual blocks enables the network to effectively model the spatial context of images. Res2Net blocks capture and propagate relevant information from earlier layers, enabling the network to exploit spatial context for denoising. An adaptive coordinate attention mechanism further optimizes this contextual information by focusing on specific locations and guiding the denoising process with spatially aware attention weights. By adjusting the image resolution in combination with down-sampling, feature combination can be performed at different image scales, which improves the robustness to scale changes and thus improves the denoising performance of the model.

D. Wavelet Attention Residual Decoder

The wavelet attention residual decoder proposed in this paper is mainly realized by up-sampling operation, convolutional layer, global and local residual blocks, and wavelet attention mechanism. This process is the opposite process to the encoding stage. The structural illustration is shown in Fig. 5. Specifically, the feature map from FGE will be sent to the decoding stage correspondingly. This process is realized by skipping the connection. The advantage of this is that the feature information extracted in the encoding stage can be provided to the decoding stage, reducing the loss of feature information loss. Up-sampling is realized by bilinear interpolation. During the up-sampling process, the resolution of the image is continuously expanded and finally returns to the same resolution as the original noise image. Skip connection combined with up-sampling can make the noise feature map restored in the decoding stage more accurate.

In the decoding stage, the further extraction and processing of feature information are mainly implemented by Res-WA, which is the wavelet attention residual block. The specific

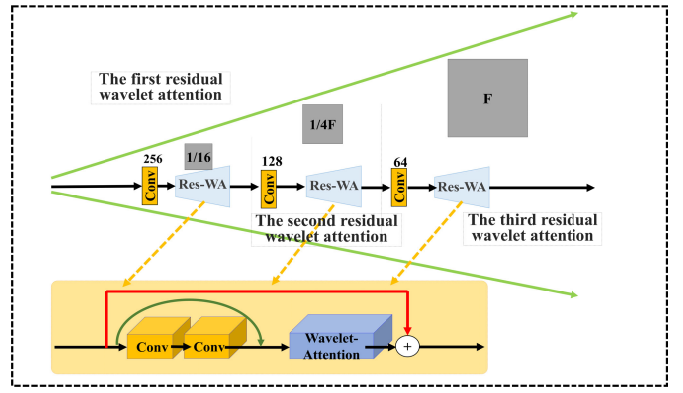


Fig. 5. Illustration of the detailed architecture of WARD. WARD comprises three iterations of residual wavelet attention reconstruction processing, which is achieved through global-local residuals and WA. During this process, the resolution of the feature maps gradually increases.

structure is depicted in the illustration of Fig. 5. Inspired by the wavelet transform and its application in the image field, we propose a wavelet attention mechanism, its novelty lies in frequency domain information to weight the feature channels, allowing for more effective preservation of edge information during the denoising process. Specifically, the channel feature maps from the previous network stage are decomposed using the discrete wavelet transform, with a one-level decomposition using Daubechies wavelets. Daubechies wavelets generate three high-frequency subbands and one low-frequency subband, emphasizing edge details in the horizontal, vertical, and diagonal directions. Zero-padding is used to retain more boundary information. In the thresholding step, learnable thresholds are set, allowing for smoother noise removal without significant loss of detail. Using normalization to maintain parameter stability. Subsequently, the three high-frequency subbands are concatenated and fused, followed by convolutional operations to further learn edge details. Finally, an inverse discrete wavelet transform is applied to reconstruct the channel feature maps.

Subsequently, the main features are extracted by employing global average pooling (GAP) to compress the feature maps. Then, the features are expanded using Flatten, and finally, wavelet attention-guided weights are generated through the Softmax function. The wavelet attention mechanism can guide the feature learning process and better preserve image edge information during the noise removal process. The specific structure of wavelet attention is depicted in Fig. 1. The working process of the wavelet attention mechanism can be described as follows.

$$\begin{cases} (H_1, H_2, H_3, L) = DWT(X, \text{Daubechies}) \\ H_{\text{concat}} = \text{Concatenate}(H_1, H_2, H_3) \\ X_{\text{re}} = \text{IDWT}(L, \text{Convolution}(H_{\text{concat}}, \text{Daubechies})) \\ \text{Final_features} = \text{Softmax}(\text{Flatten}(\text{GAP}(X_{\text{re}}))) \times X, \end{cases} \quad (4)$$

where X is the original input feature map. (H_1, H_2, H_3) are the high-frequency subbands obtained from the DWT . L represents the low-frequency subband from the DWT of X . Daubechies refers to the specific wavelet function used in DWT and $IDWT$.

In practical usage, the integration of wavelet attention with a residual network enhances noise removal. The WARD handling procedure can be summed up as follows:

$$O_{WARD} = f_{R3} \{ f_{I3} [f_{R2} (f_{I2} \{ f_{R1} [f_{I1} (O_{FGE})] \})] \} \}, \quad (5)$$

where O_{FGE} is the output of FGE, f_{I1} , f_{I2} , and f_{I3} are convolution functions that respectively represent the number of channels changed from 256 to 128, 128 to 64, and 64 to 64, f_{R1} , f_{R2} , and f_{R3} are Res-WA blocks and O_{WARD} is the output of WARD.

E. Noise Estimated Block

NEB is designed to extract a global noise map while helping to enhance denoising performance. As shown in Fig. 1, NEB consists of two parts, a 7-layer neural network composed of convolutions and dilated convolutions and a residual connection. The input of NEB is the original noisy image. Different from the backbone denoising network composed of an encoder and decoder, NEB pays more attention to modeling and estimating noise. The convolution part in NEB uses a combination of Conv, BN, and PReLU to fully extract feature information while avoiding difficult training situations. The second convolution uses dilated convolution to further increase the receptive field. In addition, residual connections reduce long-distance dependencies. Fusing the output of the noise estimation block with the output of the backbone denoising network can take full advantage of the advantages of both. The backbone denoising network focuses on recovering clean images from noisy input images, while the noise estimation block provides more accurate noise estimates. By fusing the outputs of the two, the backbone denoising network can be more targeted to remove noise, which can help the network to better adapt to different types and intensities of noise, making it more widely applicable. Moreover, by combining the outputs of the noise estimation block and the main denoising network, finer image restoration can be performed based on prior noise information. The final output of NEB is the noise estimation map. The specific process of NEB is as follows.

$$O_{NEB} = f_{C2-C6} (f_{C1D1}(N)) + f_{C1D1}(N), \quad (6)$$

where f_{C1D1} represents the first convolutional layer and the first dilated convolutional layer (second layer) of NEB, f_{C2-C6} is next five layers of convolution, $f_{C2-C6} (f_{C1D1}(N)) + f_{C1D1}(N)$ represents the residual connection and O_{NEB} is the output of the NEB.

F. Fused Reconstruction Block

In FRB, the fusion process begins by fully integrating the outputs from the encoder-decoder (which are the channel noise maps) and the NEB (which are the noise estimate maps) using concatenation and then altering the channels through convolution to create the ultimate merged noise map. Finally, a residual learning strategy is employed to obtain a clean image by subtracting the merged noise map from the input of WACAFRN (the noisy image).

The following equation can be used to describe this process:

$$C = N - C_1 (\text{Concat} (O_{WARD}, O_{NEB})), \quad (7)$$

where N represents the noisy input image, *Concat* stands for the function of fusing two noise maps, C_1 represents convolution applied to the merged feature map, and C represents the clean image predicted by the model.

IV. EXPERIMENTS AND ANALYSES

A. Experimental Setup

1) *Training Datasets*: The experiments in this study used three main training datasets. Specifically, for denoising grayscale images, we use 400 images from the BSD400 [38] dataset, the first 100 images from the Waterloo Exploration Database [39] dataset, and 800 images from the 4x enlarged version of the DIV2K [40] dataset to compose a synthetic dataset. To ensure that there are enough samples for model training, we perform data augmentation. First, we apply a bicubic interpolation method with reduction factors of 1.0, 0.9, 0.8, and 0.7 to expand the size of the training dataset, and then crop the image patch, the size of the cropped image patch is set to 40×40 , we set the training batch size to 128, resulting in a total of 17479×128 image patches being cropped for training the model.

For color image denoising, we use 3000 color images from the Waterloo Exploration Database as training images. We perform the same data augmentation method as grayscale image denoising and we cropped a total of 7356×128 image patches to train the model.

The Waterloo Exploration Database dataset and 100 JPEG-compressed 512×512 data from the Nam [41] dataset are used to create the real noisy image training dataset in addition to the synthetic noisy training dataset. Here, all the images are equally divided into 50×50 -pixel blocks for processing.

2) *Testing Datasets*: The denoising effect of WACAFRN was tested in our experiments using a total of seven test sets: BSD68 [42], Set12 [43], CBSD68 [42], Kodak24 [44], McMaster [45], CC [46] and Nam. They include BSD68 and Set12 for testing grayscale images, CBSD68, Kodak24, and McMaster for testing color images, and CC and Nam for testing images with real noise.

3) *Evaluation Metrics*: Two metrics, Peak Signal-to-Noise Ratio (PSNR) [47] and Structural Similarity (SSIM) [48], are utilized to evaluate the image denoising model's performance in this study. Higher values for both metrics indicate better image quality.

4) *Implementation Details and Optimization*: The NVIDIA Tesla V100 (32G) GPU is used in this study's experiments. Running memory is 40GB. The batch size is set to 128 and the training epoch to 100. The initial learning rate is $1e-3$; after 30 epochs, it decays to $1e-4$; and after 60 epochs, it decays to $1e-5$. The optimization method uses the Adam algorithm, and like most advanced methods [19], [23], [28], we use MSE as the loss function without introducing any additional regularization terms.

B. Experimental Analysis on Grayscale Image Denoising

For grayscale image denoising, we evaluate our proposed WACAFRN model on two common test datasets, namely Set12 and BSD68, to be clear, the test and training sets do not

TABLE I
COMPARISON OF GRAYSCALE IMAGE DENOISING RESULTS, WITH AVERAGE PSNR AND SSIM ON THE BSD68 AND SET12 DATASETS,
WHERE HIGHER VALUES INDICATE BETTER QUALITY

Methods	Model Size	$\sigma=15$ BSD68	$\sigma=15$ Set12	$\sigma=25$ BSD68	$\sigma=25$ Set12	$\sigma=50$ BSD68	$\sigma=50$ Set12
BM3D [13]	-	31.07/0.8719	32.37/0.8948	28.57/0.8012	29.97/0.8493	25.62/0.6867	26.72/0.7664
WNNM [12]	-	31.37/0.8766	32.70/0.8982	28.83/0.8087	30.26/0.8557	25.87/0.6982	27.05/0.7775
EPLL [18]	-	31.21/-	32.14/-	28.68/-	29.69/-	25.67/-	26.47/-
MLP [15]	-	-	-	28.96/-	30.03/-	26.03/-	26.78/-
CSF [16]	-	31.24/-	32.32/-	28.74/-	29.84/-	-	-
TNRD [17]	0.03M	31.42/0.8788	32.50/0.8920	28.92/0.8019	30.06/0.8497	25.97/0.7012	26.81/0.7687
DnCNN [19]	0.56M	31.73/0.8907	32.86/0.9024	29.23/0.8279	30.43/0.8617	26.23/0.7189	27.18/0.7828
IRCNN [21]	4.45M	31.63/0.8882	32.77/0.9006	29.15/0.8248	30.38/0.8598	26.19/0.7169	27.14/0.7804
FFDNet [20]	0.85M	31.62/0.8901	32.77/0.9025	29.19/0.8288	30.44/0.8632	26.29/0.7239	27.32/0.7899
ECNDNet [49]	0.50M	31.71/-	32.81/-	29.22/-	30.39/-	26.23/-	27.15/-
RIDNet [22]	1.50M	31.81/0.8926	32.91/0.9030	29.34/0.8310	30.60/0.8678	26.40/0.7246	27.43/0.7930
ADNet [23]	0.52M	31.74/0.8916	32.98/0.9049	29.25/0.8294	30.58/0.8654	26.29/0.7216	27.37/0.7898
AINNet [50]	13.76M	31.69/0.8865	32.92/0.9028	29.26/0.8302	30.60/0.8641	26.32/0.7173	27.43/0.7927
DudeNet [24]	1.03M	31.78/0.8918	32.94/0.9036	29.29/0.8294	30.53/0.8633	26.31/0.7167	27.30/0.7839
DBDNet [25]	4.45M	31.85/0.8942	33.03/0.9055	29.37/0.8321	30.66/0.8662	26.43/0.7274	27.48/0.7932
COLA-Net [51]	1.10M	31.88/0.8938	33.20/0.9088	<u>29.42/0.8339</u>	<u>30.85/0.8701</u>	<u>26.48/0.7308</u>	<u>27.73/0.8020</u>
WINNet [32]	0.35M	31.72/-	32.86/-	29.24/-	30.54/-	26.36/-	27.40/-
MWDCNN [28]	5.01M	31.77/0.8915	32.91/0.9028	29.28/0.8291	30.55/0.8637	26.29/0.7160	27.34/0.7852
WACAFRN	2.43M	31.90/0.8949	33.20/0.9103	29.43/0.8349	30.86/0.8792	26.51/0.7322	27.77/0.8036

Note: The best results for each test dataset with each noise level are **bold**, and the second results are underlined.

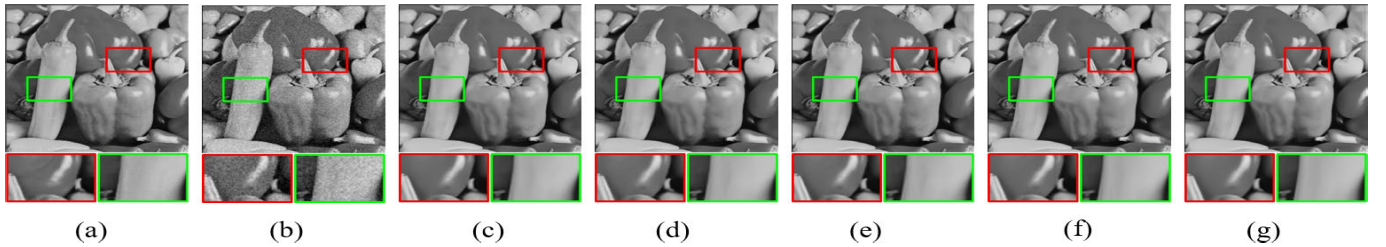


Fig. 6. Comparison of denoising performance of different methods on the same image from Set12, $\sigma = 15$: (a) original image, (b) noisy image, (c) DnCNN (PSNR=33.24dB), (d) IRCNN (PSNR=33.21dB), (e) FFDNet (PSNR=33.01dB), (f) DudeNet (PSNR=33.38 dB), (g) WACAFRN (PSNR=33.52 dB).

intersect. We compare WACAFRN with 18 state-of-the-art image denoising models, including traditional classical image denoising methods such as WNNM [12], BM3D [13], and EPLL [18], as well as discriminative and deep learning methods, namely MLP [15], CSF [16], TRND [17], DnCNN [19], IRCNN [21], FFDNet [20], ECNDNet [49], RIDNet [22], ADNet [23], AINNet [50], DudeNet [24], DBDNet [25], COLA-Net [51], WINNet [32], and MWDCNN [28]. To logically confirm the performance of our suggested technique, we select representative noise cases, namely $\sigma = 15, 25$, and 50 , from three levels: low noise, medium noise, and high noise for comparative experiments. Table I shows the average PSNR and SSIM comparison of various models on BSD68 and Set12 under $\sigma = 15, 25, 50$. According to Table I, our proposed WACAFRN achieves the best PSNR and SSIM values at each of the three noise levels. On two datasets, as shown in Table I, WACAFRN gets the best structural similarity under three noise levels, proving that it is more capable of repairing the intricate structural details of images. We visualized denoising results to compare various methods. Fig. 6 highlights the results of WACAFRN and other models on the Set12 dataset at noise level 15. Key details in the image are encircled and enlarged twice for clarity. The WACAFRN's result closely matches the

original image's contour, indicating its superior visual effect compared to other models.

C. Experimental Analysis on Color Image Denoising

We compared seven advanced methods for the task of color image denoising. The comparison was conducted on three public test datasets, considering five different noise levels. According to the results presented in Table II, WACAFRN exhibited the best denoising performance across all test examples. Particularly, WACAFRN outperformed the alternative methods, showcasing its superior noise resistance capabilities, especially in high noise level scenarios. In Fig. 7, we provided a visual comparison of the denoising effects of each method on an image from the Kodak24 dataset. To emphasize image details, we framed and magnified the circled regions. Upon visual inspection, it was evident that IRCNN resulted in image blurring, while DudeNet introduced noticeable line deformation. In contrast, WACAFRN consistently achieved superior visual results.

D. Experimental Analysis on Real-World Denoising

While synthetic noise research can support and validate denoising model performance, real-world noise is typically

TABLE II

COMPARISON OF DENOISING RESULTS FOR COLOR IMAGES, WITH AVERAGE PSNR ON THE CBSD68, KODAK24, AND MCMASTER DATASETS, WHERE HIGHER VALUES INDICATE BETTER DENOISING PERFORMANCE

Datasets	Noise Level	CBM3D	DnCNN	FFDNet	IRCNN	ADNet	DudeNet	MWDCNN	WACAFRN
CBSD68	15	33.52	33.98	33.80	33.86	33.99	34.01	34.18	34.29
	25	30.71	31.31	31.18	31.16	31.31	31.34	31.45	31.51
	35	28.89	29.65	29.57	29.50	29.66	29.71	29.81	29.96
	50	27.38	28.01	27.96	27.86	28.04	28.09	28.13	28.19
	75	25.74	-	26.24	-	26.33	26.40	26.40	26.47
Kodak24	15	34.28	34.73	34.55	34.56	34.76	34.81	34.92	34.98
	25	31.68	32.23	32.11	32.03	32.26	32.26	32.38	32.45
	35	29.90	30.64	30.56	30.43	30.68	30.69	30.88	30.93
	50	28.46	29.02	28.99	28.81	29.10	29.10	29.27	29.32
	75	26.82	-	27.27	-	27.40	27.39	27.57	27.61
McMaster	15	34.06	34.80	34.47	34.58	34.93	34.83	34.86	35.15
	25	31.66	32.47	32.25	32.18	32.56	32.40	32.52	32.76
	35	29.92	30.91	30.76	30.59	31.00	30.98	31.05	31.12
	50	28.51	29.21	29.14	28.91	29.36	29.12	-	29.60
	75	26.79	27.33	25.10	27.36	27.53	-	-	28.02

Note: The best results for each test dataset with each noise level are **bold**.

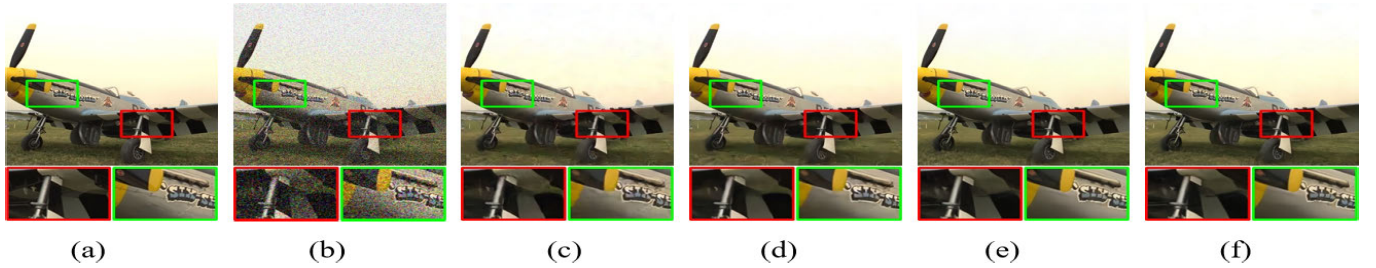


Fig. 7. Comparison of denoising effects of different methods on the same image from Kodak24, $\sigma = 50$: (a) original image (b) noisy image (c) IRCNN (PSNR=30.89dB), (d) ADNet (PSNR=30.96dB), (e) DudeNet (PSNR=31.12dB), (f) WACAFRN (PSNR=31.54dB).

complex, diverse, and unpredictable. It encompasses various environmental factors, sensor characteristics, and image conditions. Experimenting with real-world noise allows us to assess denoising methods in realistic scenarios, understand their strengths and limitations, and evaluate their robustness and generalizability across different noise types, intensities, and image characteristics. We conducted experiments using the WACAFRN method for real-world noise removal on the CC and Nam datasets. On the CC dataset, we compared WACAFRN with five advanced methods. As shown in Table III, WACAFRN consistently outperformed all other methods, achieving the highest PSNR and SSIM values in all 15 scenarios and the highest average PSNR and SSIM result, surpassing other methods by at least 2.08 dB in PSNR. Notably, WACAFRN achieved PSNR values of 40.00 dB or higher in multiple cases, indicating high-quality denoised images. On the Nam dataset, we compared WACAFRN with BM3D, MLP, DnCNN, FFDNet, RIDNet, CBDNet, SADNet [53], MPRNet [52], AINDNet, and MIRNet [54]. Based on the results presented in Table IV, our proposed method achieved the best PSNR and SSIM results, specifically measuring 40.53 dB and 0.989, respectively. Considering the results on both datasets, it's evident that our proposed method can achieve outstanding performance with effective

computational complexity in real-world noise removal tasks.

E. Ablation Study

In this section, we delve into the significance and reasoning behind each module in WACAFRN through ablation experiments. These experiments targeted real-world noise removal on the CC test set and Gaussian denoising tasks at noise levels of 50 on the BSD68 dataset. For a comparative perspective, we set WACAFRN as our baseline. The study removed each module systematically to see its impact on PSNR and model parameters. The outcomes are presented in Tables V and VI, offering insights into each module's role.

To demonstrate the impact of removing ACA (with 0.02M parameters), we conducted experiments. As a result, on the CC dataset, PSNR decreased from 37.82 to 37.19. This indicates that ACA plays a positive role in capturing and more thoroughly removing noise, and our improved ACA outperforms the original CA attention.

To validate the effectiveness of the wavelet attention mechanism (with 0.12M parameters), we removed it, and by observing the experimental results, it is evident that wavelet attention plays a crucial role in improving model performance. To further validate the contribution of WA in preserving image

TABLE III

COMPARISON OF THE AVERAGE PSNR/SSIM VALUES FOR DENOISED REAL IMAGES ON THE CC DATASET. HIGHER VALUES FOR BOTH PSNR AND SSIM INDICATE BETTER DENOISING PERFORMANCE

Dataset settings	CBM3D	DnCNN	ADNet	DudeNet	MWDCNN	WACAFRN
Nikon-d600 Image_3200_1	34.18/0.9330	33.62/0.9370	33.94/0.9452	33.72/0.9381	33.91/0.9497	35.47/0.9565
Nikon-d600 Image_3200_2	35.07/0.9168	34.48/0.9182	34.33/0.9110	34.70/0.9245	34.88/0.9302	37.12/0.9557
Nikon-d600 Image_3200_3	37.13/0.9313	35.41/0.9362	38.87/0.9351	37.98/0.9364	37.02/0.9370	40.35/0.9840
Nikon-d800 Image_1600_1	36.81/0.9339	37.95/0.9575	37.61/0.9458	38.10/0.9591	37.93/0.9572	39.33/0.9681
Nikon-d800 Image_1600_2	37.76/0.9383	36.08/0.9320	38.24/0.9601	39.15/0.9684	37.49/0.9359	41.82/0.9812
Nikon-d800 Image_1600_3	37.51/0.9277	35.48/0.9230	36.89/0.9341	36.14/0.9311	38.44/0.9416	39.23/0.9537
Nikon-d800 Image_3200_1	35.05/0.8866	34.08/0.8623	37.20/0.8969	36.93/0.9005	37.10/0.9115	40.28/0.9772
Nikon-d800 Image_3200_2	34.07/0.8928	33.70/0.8852	35.67/0.9195	35.80/0.9260	36.72/0.9356	37.65/0.9575
Nikon-d800 Image_3200_3	34.42/0.8430	33.31/0.8359	38.09/0.9465	37.49/0.9401	37.25/0.9475	41.35/0.9733
Nikon-d800 Image_6400_1	31.13/0.7952	29.83/0.7611	32.24/0.7994	31.94/0.8210	32.24/0.8547	33.62/0.9127
Nikon-d800 Image_6400_2	31.22/0.8613	30.55/0.8598	32.59/0.9141	32.51/0.9153	32.56/0.9177	33.51/0.9341
Nikon-d800 Image_6400_3	30.97/0.8363	30.09/0.7941	33.14/0.9105	32.91/0.8947	32.76/0.8988	33.25/0.9178
Canon-5d Image_3200_1	39.76/0.9778	37.26/0.9623	35.69/0.9630	36.66/0.9702	36.97/0.9725	40.12/0.9819
Canon-5d Image_3200_2	36.40/0.9550	34.87/0.9467	36.11/0.9398	36.70/0.9477	36.01/0.9491	37.18/0.9602
Canon-5d Image_3200_3	36.37/0.9662	34.09/0.9451	34.49/0.9475	35.03/0.9509	34.80/0.9478	37.03/0.9695
Average	35.19/0.9063	34.05/0.8971	35.67/0.9246	35.72/0.9283	35.74/0.9325	37.82/0.9589

Note: The best results are **bold**.

TABLE IV

COMPARISON OF THE AVERAGE PSNR/SSIM VALUES FOR DENOISED REAL IMAGES ON THE NAM DATASET. HIGHER VALUES FOR BOTH PSNR AND SSIM INDICATE BETTER DENOISING PERFORMANCE. THE DENOISING INFERENCE TIME IS THE AVERAGE TIME OBTAINED WITH AN IMAGE SIZE OF 512×512

Methods	Time (s)	Parameters	PSNR	SSIM
BM3D [13]	-	-	39.84	0.965
MLP [15]	-	-	34.63	0.852
DnCNN [19]	0.027	0.56M	34.95	0.885
FFDNet [20]	0.014	0.85M	38.81	0.960
RIDNet [22]	0.143	1.50M	39.20	0.973
CBDNet [35]	0.392	6.79M	40.02	0.969
MPRNet [52]	0.882	15.74M	39.97	0.987
SADNet [53]	0.256	4.77M	39.89	0.979
AINDNet [50]	0.319	13.76M	39.09	0.962
MIRNet [54]	1.735	31.78M	40.15	0.980
WACAFRN	0.039	2.43M	40.53	0.989

Note: The best results are **bold**.

TABLE V

ABLATION STUDY OF KEY COMPONENTS ON THE CC DATASET

Model	PSNR(dB)	Parameters
WACAFRN	37.82	2.43M
WACAFRN without NEB	37.62	2.25M
NEB without dilated convolution	37.67	2.36M
WACAFRN without ACA	37.19	2.41M
WACAFRN with original CA	37.24	2.42M
WACAFRN without WA	36.95	2.31M
WACAFRN with only one Res2Net	37.22	2.09M
WACAFRN without stacked Res2Net	36.87	1.92M
WACAFRN without decoding stage	36.55	1.78M
WACAFRN without PFEB	37.59	2.29M

TABLE VI

ABLATION STUDY OF KEY COMPONENTS ON THE BSD68 DATASET WITH NOISE LEVEL 50

Model	PSNR(dB)	Parameters
WACAFRN	26.51	2.43M
WACAFRN without NEB	26.46	2.25M
NEB without dilated convolution	26.48	2.36M
WACAFRN without ACA	26.41	2.41M
WACAFRN with original CA	26.44	2.42M
WACAFRN without WA	26.38	2.31M
WACAFRN with only one Res2Net	26.44	2.09M
WACAFRN without stacked Res2Net	26.37	1.92M
WACAFRN without decoding stage	26.31	1.78M
WACAFRN without PFEB	26.48	2.29M

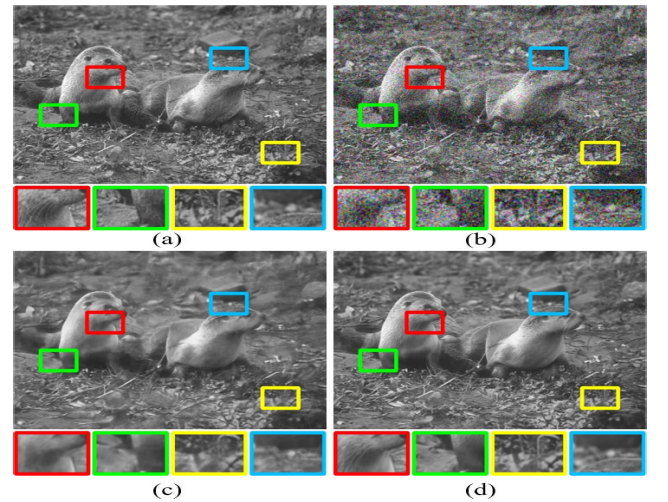


Fig. 8. Ablation comparison for the wavelet attention mechanism, comparing denoising performance on the same image (from BSD68) at a noise level $\sigma = 50$: (a) original image, (b) noisy image, (c) denoised result by WACAFRN without wavelet attention, (d) denoised result by WACAFRN.

edge information, we compared the denoising results with and without WA, as shown in Fig. 8. We selected four detailed regions in the same image with rich edge transformations and

magnified them by a factor of two. By visualizing the results, WA effectively preserves image edge information during the denoising process.

The ablation study on other parts of the model primarily focused on the following aspects: Firstly, to validate the role of stacked fine-grained networks (0.51M), we conducted comparative experiments by removing stacked Res2Net blocks and removing only one Res2Net block. Next, we verified the role of PFEB (0.14M) by removing it from the model. The results showed a significant impact on the model's performance, demonstrating the contribution of PFEB. Finally, we conducted comparative experiments by removing NEB (0.18M) to validate the performance enhancement achieved by incorporating noise estimation information.

V. CONCLUSION

In this study, a fine-grained residual denoising network guided by wavelet and adaptive coordinate attention is proposed. Unlike most deep neural networks, the design of WACAFRN stems from better acquisition and utilization of noise channel information and position information, improving the reasoning ability of the model, and better understanding and restoring image edge information. WACAFRN consists of five main parts: PFEB, FGE, WARD, NEB, and FRB, each working in coordination with the other. The influence of key blocks on the image denoising effect is verified by ablation experiments. Extensive experiments show that our proposed WACAFRN can achieve competitive performance in the tasks of grayscale image denoising, color image denoising, and real-world image denoising. In the future, considering the complexity and diversity of real-world noise, we are committed to continuing to improve the model and further improve the effect of real-world denoising.

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